

RL Assignment 1 problem 4

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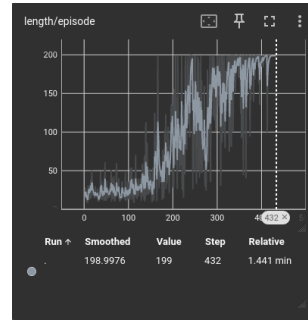
1 (a)

For the actor and the value agent, i design a MLP with shared layer and each output layer, the shared layer is a two full-connected layer with ReLU() activation function. The parameter is: hidden size 1=128, hidden size 2 = 64. And we initialize the for the linear layer's weight with mean = 0, and std = 0.01, in case of some action is closed to 1 at the beginning.

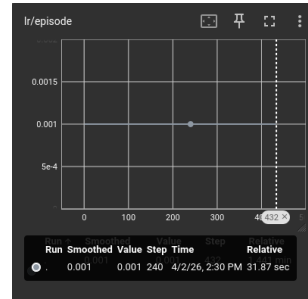
As for the calculate loss function, we first normalize the returns, because in the Cart-Pole problem, the rewards are always positive, the model will increase the probability of all actions, just differ by the amount, but with the normalization, the Return value have positive and negative number, helping the model to learn the policy. As for the loss, loss equals to policy loss plus value loss, the policy loss is just substitute in REINFORCE policy gradient, as for the value loss, we use return as the estimator to estimate the value function, so i use MSE to estimate the loss between value function and return. The training part, i pick learning rate equals to 0.001.

1.1 results

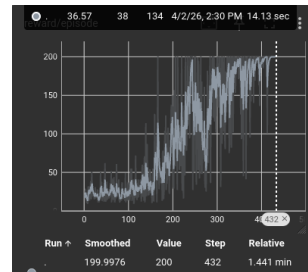
It takes about 432 steps to converge, and qualify the threshold of this problem.



(a) length



(b) lr



(c) reward

Figure 1: Caption

2 (b)

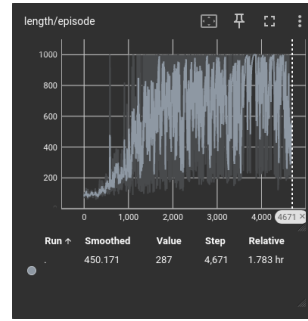
For problem b, we change the problem from Cart-Pole problem to LunarLander- v3 problem, in this problem , i use value function as baseline function to the policy gradient equation. But at the training time, i found out that the policy loss and Value loss have a big difference in their value, if we don't use coefficient to lower the value loss, the optimizer will take most weight to update the value function prediction error instead of finding better policy. And i add Gradient clipping in case of the massive gradient happening for enhancing the training stability.

2.1 results

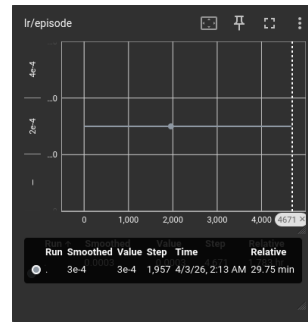
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Solved! Running reward is now 201.11885191981625 and the last episode runs to 287 time steps!
Episode 1   Reward: 283.98833993191863
Episode 2   Reward: 226.11316044280414
Episode 3   Reward: 277.5096958086415
Episode 4   Reward: 238.9660992841853
Episode 5   Reward: 244.09222928313594
Episode 6   Reward: 252.53233272373527
Episode 7   Reward: 277.675047470228
Episode 8   Reward: -2.728746471908508
Episode 9   Reward: 271.29860377406374
Episode 10  Reward: 201.8098531134717
```

Figure 2: Caption

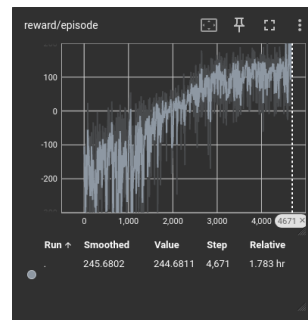
It takes about 4671 steps to converge and qualify the threshold of this problem.



(a) length



(b) lr



(c) reward

Figure 3: Caption

3 (c)

We also use LunarLander-v3 problem. But in this case, instead of using returns minus value function in the policy gradient, we use GAE to calculate the advantage function and substitute in policy gradient. But while i'm training, it usually stuck at the fixed place and never learn better policy, in every episode, the length equals 999, and the reward are stuck in -20 to -90, means the agent is in local optimal choice. In this case, i use entropy to encourage exploration to help the policy from sticking in the local optimal policy. Due to the time issue, i only do 1 $\lambda = 0.95$.

3.1 results

Run 10000 steps but still can't converge. I try the entropy method but still don't work.